

Module 10: Statistical Inference

Instructional Hours: 4

Statistical Inference

Having recalled the concepts you learnt in Module 9, let us now introduce this Module by understanding some simple concepts in Statistical Inference as explained in the videos attached below. In each video, attempt the question that follow.

Statistic vs Parameter and Population vs Sample

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/Mb9BuEkbaHQ>



Statistic vs Parameter and Population vs Sample

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/MiuDxyVmlwg>



Point Estimation and Interval Estimation

Statistical Estimation Techniques

Recall: Parameter Estimation involves using sample data to estimate the population parameters. Estimates of population parameters can be obtained using either point estimation or interval estimation methods.

Point Estimation

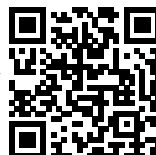
In point estimation, you obtain a single value that serves as the best estimate of an unknown population parameter. This single value is known as point estimate.

Conventionally, if θ is the parameter of interest, then $\hat{\theta}$ is the point estimate.

The main objective in point estimation is obtaining a "good" estimate of θ based on some sample data. This means that point estimators must possess some desirable qualities. You can understand this by watching the video clip below:

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/ZxUIHXIggfU>



Properties of Good Estimators

- From the video you have just watched, what are the main properties of good estimators? In particular, how could you evaluate efficiency of an estimator?

Properties of Point Estimators

A good point estimator satisfies the following properties:

1. **Unbiasedness:** An estimator $\hat{\theta}$ is unbiased if the expected value of $\hat{\theta}$ equals the true parameter θ , i.e.,

$$\mathbb{E}[\hat{\theta}] = \theta$$

That is, an unbiased estimator, on average, gives the correct parameter value.

2. **Consistency:** An estimator is consistent if, as the sample gets large, the estimator converges to the true parameter value. This property can be expressed mathematically, as $n \rightarrow \infty$,

$$\hat{\theta} \xrightarrow{P} \theta.$$

3. **Efficiency:** Among unbiased estimators, an efficient estimator has the smallest variance. That is, for any two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$, $\hat{\theta}_1$ is more efficient if

$$\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$$

4. **Sufficiency:** An estimator is sufficient if it captures all the information from the sample that is relevant to estimating the parameter θ . A sufficient statistic provides a complete summary of the data regarding the parameter of interest.

Common Point Estimation Techniques

Commonly used methods in point estimation include:

Method of Moments (MoM)

Method of moments is based on the principle of equating sample moments to population moments. A moment is the expected value of a function of the data. For instance, the first moment about the origin is the mean, the second central moment is the variance, and so on. For example, to estimate the population mean μ using the first sample moment (sample mean), we set the sample mean equal to the population mean:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Higher-order moments can be used to estimate additional parameters such as the variance.

Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation (MLE), which is a popular and powerful method of estimation, involves finding the parameter values that maximize the likelihood function, which represents the probability of observing the given sample. If the data consist of independent observations $x_1, x_2, \dots, x_n$ from a probability distribution with parameter θ , the likelihood function is given by

$$L(\theta \mid x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta)$$

where $f(x_i; \theta)$ is the probability density function (or probability mass function) of the data point x_i .

The maximum likelihood estimate $\hat{\theta}$ is the value of θ that maximizes $L(\theta)$.

It is often convenient to work with the natural logarithm of the likelihood function, known as the log-likelihood function:

$$\ell(\theta) = \log L(\theta)$$

The MLE is found by solving the equation

$$\frac{d\ell(\theta)}{d\theta} = 0$$

Bayesian Estimation

In Bayesian estimation, parameters are treated as random variables with their own probability distributions. The goal is to derive the posterior distribution of the parameter θ given the data, using Bayes' theorem:

$$p(\boldsymbol{\theta} \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{x})}$$

where $p(\boldsymbol{\theta})$ is the prior distribution, $p(\mathbf{x} \mid \boldsymbol{\theta})$ is the likelihood, and $p(\mathbf{x})$ is the marginal likelihood (a normalizing constant). The posterior distribution incorporates both prior information and the observed data.

The Bayesian point estimate is often taken as the mean or mode of the posterior distribution:

$$\hat{\boldsymbol{\theta}}_{\text{Bayesian}} = \mathbb{E}[\boldsymbol{\theta} \mid \mathbf{x}].$$

Illustration of Point Estimation for Normal Distribution

Suppose we have a sample $x_1, x_2, \dots, x_n$ drawn from a normal distribution $N(\mu, \sigma^2)$, where μ is the unknown mean and σ^2 is the unknown variance. We can estimate μ and σ^2 as follows:

- **Sample Mean:** The sample mean $\bar{x}$ is the maximum likelihood estimator (MLE) and the method of moments estimator for the population mean μ . It is given by:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i.$$

- **Sample Variance:** The sample variance S^2 is the estimator for the population variance σ^2 . The unbiased estimator for σ^2 is:

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Worked Out Examples

Example 1. Problem 1: Estimating the Mean of a Normal Distribution Suppose we have a random sample of size $n = 10$ from a normal distribution $N(\mu, \sigma^2)$ with unknown mean μ and known variance $\sigma^2 = 25$. The sample observations are as follows:

$$x_1 = 7, \quad x_2 = 10, \quad x_3 = 6, \quad x_4 = 9, \quad x_5 = 8, \quad x_6 = 11,$$

$$x_7 = 10, \quad x_8 = 12, \quad x_9 = 13, \quad x_{10} = 9.$$

Estimate the population mean μ using the method of moments and the maximum likelihood estimation (MLE).

Solution

For both the method of moments and MLE, the estimator for the population mean μ is the sample mean $\bar{x}$.

The sample mean is calculated as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{7 + 10 + 6 + 9 + 8 + 11 + 10 + 12 + 13 + 9}{10} = \frac{95}{10} = 9.5.$$

Thus, the point estimate of the population mean μ is:

$$\hat{\mu} = 9.5$$

Example 2. Problem 2: Estimating the Variance of a Normal Distribution

Consider a random sample from a normal distribution $N(\mu, \sigma^2)$ with unknown variance σ^2 and known mean $\mu = 50$. The sample consists of $n = 5$ observations:

$$x_1 = 48, \quad x_2 = 53, \quad x_3 = 50, \quad x_4 = 52, \quad x_5 = 47.$$

Estimate the population variance σ^2 using the unbiased estimator for the sample variance.

Solution

The unbiased estimator for the population variance σ^2 is the sample variance S^2 , given by:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

where $\bar{x}$ is the sample mean.

First, calculate the sample mean:

$$\bar{x} = \frac{48 + 53 + 50 + 52 + 47}{5} = \frac{250}{5} = 50.$$

Now calculate the squared deviations from the mean:

$$\begin{aligned} (x_1 - \bar{x})^2 &= (48 - 50)^2 = 4, & (x_2 - \bar{x})^2 &= (53 - 50)^2 = 9, \\ (x_3 - \bar{x})^2 &= (50 - 50)^2 = 0, & (x_4 - \bar{x})^2 &= (52 - 50)^2 = 4, \\ (x_5 - \bar{x})^2 &= (47 - 50)^2 = 9. \end{aligned}$$

Summing these squared deviations gives:

$$\sum_{i=1}^n (x_i - \bar{x})^2 = 4 + 9 + 0 + 4 + 9 = 26$$

Finally, calculate the sample variance:

$$S^2 = \frac{26}{5-1} = \frac{26}{4} = 6.5$$

Thus, the point estimate for the population variance σ^2 is:

$$\hat{\sigma}^2 = 6.5$$

Example 3. Problem 3: Maximum Likelihood Estimation for Exponential Distribution

Let $X_1, X_2, \dots, X_n$ be a random sample from an exponential distribution with the probability density function (pdf):

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0$$

where $\lambda > 0$ is the unknown rate parameter. Given a sample $x_1 = 2, x_2 = 1, x_3 = 3, x_4 = 2$, find the maximum likelihood estimate (MLE) of λ .

Solution

The likelihood function for the sample is:

$$L(\lambda \mid x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i; \lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

The log-likelihood function is:

$$\ell(\lambda) = \log L(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i$$

To find the MLE of λ , we take the derivative of the log-likelihood function with respect to λ and set it to zero:

$$\frac{d\ell(\lambda)}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0$$

Solving for λ :

$$\hat{\lambda} = \frac{n}{\sum_{i=1}^n x_i}.$$

For the given sample, we have:

$$\sum_{i=1}^4 x_i = 2 + 1 + 3 + 2 = 8$$

Thus, the MLE of λ is:

$$\hat{\lambda} = \frac{4}{8} = 0.5$$

Example 4. Problem 4: Estimating the Proportion in a Bernoulli Distribution

A researcher conducts an experiment to estimate the proportion of defective items in a production line. Out of 100 sampled items, 15 are found to be defective. Estimate the population proportion p of defective items using the maximum likelihood estimation (MLE).

Solution

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables from a Bernoulli distribution with parameter p . The likelihood function for the Bernoulli distribution is:

$$L(p \mid x_1, x_2, \dots, x_n) = p^{\sum_{i=1}^n x_i} (1-p)^{n-\sum_{i=1}^n x_i}.$$

The log-likelihood function is:

$$\ell(p) = \sum_{i=1}^n x_i \log p + \left(n - \sum_{i=1}^n x_i \right) \log(1-p)$$

Taking the derivative of the log-likelihood function with respect to p and setting it to zero:

$$\frac{d\ell(p)}{dp} = \frac{\sum_{i=1}^n x_i}{p} - \frac{n - \sum_{i=1}^n x_i}{1-p} = 0$$

Solving for p , we get the MLE for the population proportion:

$$\hat{p} = \frac{\sum_{i=1}^n x_i}{n}$$

In this case, $\sum_{i=1}^n x_i = 15$ and $n = 100$, so the MLE for p is:

$$\hat{p} = \frac{15}{100} = 0.15$$

Use of R Software in Point Estimation

In this section, you will learn on how R software can be used to obtain various point estimates of unknown parameters.

Example 5. Estimating the Mean of a Normal Distribution

Consider a dataset that is generated from a normal distribution with unknown mean μ and known standard deviation $\sigma = 2$. You are given the following sample data:

$$x = [4.5, 5.0, 5.5, 6.0, 6.5] \quad (32)$$

Use R to compute the Maximum Likelihood Estimate (MLE) for the mean μ .

Solution

The MLE for the mean of a normal distribution is simply the sample mean. The R code for this question is:

```
# Sample data
x <- c(4.5, 5.0, 5.5, 6.0, 6.5)

\# Compute the MLE for the mean
mean_estimate <- mean(x)
mean_estimate
```

Example 6. Estimating the Rate Parameter of an Exponential Distribution

Suppose the time between successive events follows an exponential distribution with unknown rate parameter λ . The following waiting times (in minutes) are observed:

$$t = [1.2, 0.5, 0.8, 1.5, 2.0] \quad (33)$$

Use R to compute the MLE for the rate parameter λ .

Solution

The MLE for the rate parameter λ of an exponential distribution is given by:

$$\hat{\lambda} = \frac{1}{\bar{t}}$$

where $\bar{t}$ is the sample mean of the waiting times. We can compute this in R as follows:

```
# Waiting times
t <- c(1.2, 0.5, 0.8, 1.5, 2.0)

# Compute the MLE for lambda
```



```
lambda_estimate <- 1 / mean(t)
```

```
# Display the estimate
lambda_estimate
```

Bayesian Point Estimate for a Gaussian Distribution

Example 7. Suppose we have a prior belief that the mean of a normally distributed dataset (with known variance $\sigma^2 = 1$) is $\mu_0 = 3$ with a prior variance of $\tau^2 = 4$. You observe the following data points:

$$x = [2.8, 3.1, 3.0, 2.9, 3.2] \quad (34)$$

Compute the posterior mean of μ using Bayesian estimation in R.

Solution

For Bayesian estimation with a Gaussian prior and likelihood, the posterior mean is given by:

$$\hat{\mu}_{\text{posterior}} = \frac{\frac{1}{\sigma^2} \sum x_i + \frac{1}{\tau^2} \mu_0}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

where n is the number of data points. We can compute this in R as follows:

```
# Sample data
x <- c(2.8, 3.1, 3.0, 2.9, 3.2)

# Known parameters
mu_0 <- 3
tau2 <- 4
sigma2 <- 1
n <- length(x)

# Compute the posterior mean
posterior_mean <- (sum(x) / sigma2 + mu_0 / tau2) / (n / sigma2 + 1 / tau2)

# Display the posterior mean
posterior_mean
```

Interval Estimation

Definition. In interval estimation, you use statistical tools to obtain a range of plausible values for the parameter of interest, usually denoted as θ .

Unlike point estimation where you obtain a single estimate of the parameter, in interval estimation, you will calculate a range or interval within which an unknown population parameter is expected to lie with a certain level of confidence. This means that a Confidence Interval (CI) provides an estimate of the parameter with a specified level of confidence (e.g., 95%, 98%).

Definition. *Confidence interval for a parameter θ is an interval $[\hat{\theta}_L, \hat{\theta}_U]$ such that the true parameter lies within this interval with a specified probability, known as the confidence level.*

For instance, a 95% confidence interval for θ is given by:

$$P(\hat{\theta}_L \leq \theta \leq \hat{\theta}_U) = 0.95$$

If population μ is your parameter of interest, you can construct the confidence interval for μ based on a normal distribution as:

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right),$$

where $\bar{x}$ is the sample mean and σ is the population standard deviation, while n is the sample size and $z_{\alpha/2}$ is the critical value from the standard normal distribution corresponding to the confidence level.

Common Confidence Intervals and their Applications

The general form of a confidence interval for a population parameter θ is:

$$\text{Confidence Interval} = (\hat{\theta} - E, \hat{\theta} + E)$$

where $\hat{\theta}$ is the point estimate (usually, statistic of interest) and E is the margin of error. The margin of error depends on the standard error of the estimate and the desired level of confidence.

Usually, confidence intervals are defined by the following confidence levels:

- 90% confidence level: implies that in 90% of repeated samples, the true population parameter would fall within the interval.
- 95% confidence level: indicates that you are 95% confident that the interval contains the true parameter.
- 99% confidence level: implies a wider interval to be more certain about containing the true parameter.

General Formula for Confidence Intervals

For many common population parameters, the confidence interval can be written as:

$$\text{CI} = \hat{\theta} \pm z_{\alpha/2} \cdot \text{SE}(\hat{\theta})$$

where:

- $\hat{\theta}$ is the point estimate (e.g., sample mean $\bar{x}$).
- $z_{\alpha/2}$ is the critical value from the standard normal distribution corresponding to the confidence level.
- $\text{SE}(\hat{\theta})$ is the standard error of the point estimate.

Confidence Interval for the Population Mean

Case I: Known Population Variance

Assuming a known population variance σ^2 , the confidence interval for the population mean μ is given by:

$$\text{CI} = \bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

where:

- $\bar{x}$ is the sample mean.
- $z_{\alpha/2}$ is the critical value from the standard normal distribution.
- σ is the known population standard deviation.
- n is the sample size.

Example 8. If $\bar{x} = 100, \sigma = 15, n = 25$, and the desired confidence level is 95%, we look up $z_{0.025} = 1.96$. The confidence interval is:

$$\text{CI} = 100 \pm 1.96 \cdot \frac{15}{\sqrt{25}} = 100 \pm 5.88$$

Thus, the 95% confidence interval is (94.12, 105.88).

Case II: Unknown Population Variance

When the population variance σ^2 is unknown, you use the sample standard deviation s and the t -distribution with $n - 1$ degrees of freedom to construct the confidence interval so that the confidence interval for the population mean μ is:

$$CI = \bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}},$$

where:

- $t_{\alpha/2, n-1}$ is the critical value from the t -distribution with $n - 1$ degrees of freedom.
- s is the sample standard deviation.

Example 9. Suppose $\bar{x} = 50, s = 8, n = 16$, and the desired confidence level is 95%. We find $t_{0.025, 15} \approx 2.131$. The confidence interval is:

$$CI = 50 \pm 2.131 \cdot \frac{8}{\sqrt{16}} = 50 \pm 4.262$$

Thus, the 95% confidence interval is (45.738, 54.262).

Confidence Interval for Population Proportion

Suppose you let p to represent the proportion of successes in a population, and $\hat{p}$ represent the sample proportion. You can obtain the confidence interval for p as follows:

$$CI = \hat{p} \pm z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where:

- $\hat{p}$ is the sample proportion of successes.
- $z_{\alpha/2}$ is the critical value from the standard normal distribution.
- n is the sample size.

Example

Suppose in a sample of 200 voters, 120 say they will vote for candidate A . The sample proportion is:

$$\hat{p} = \frac{120}{200} = 0.6$$

For a 95% confidence interval, $z_{0.025} = 1.96$. The confidence interval is:

$$CI = 0.6 \pm 1.96 \cdot \sqrt{\frac{0.6(1 - 0.6)}{200}} = 0.6 \pm 0.068$$

Thus, the 95% confidence interval is (0.532, 0.668).

Confidence Interval for Population Variance

Assuming a normally distributed population, you can get the confidence interval for the variance σ^2 , based on the chi-square (χ^2) distribution. The confidence interval is:

$$CI = \left(\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2} \right),$$

where:

- s^2 is the sample variance.
- $\chi_{\alpha/2, n-1}^2$ and $\chi_{1-\alpha/2, n-1}^2$ are critical values from the chi-square distribution.

Example 10. Suppose $n = 20$, $s^2 = 25$, and we want a 95% confidence interval for the variance. Using the chi-square distribution with 19 degrees of freedom, $\chi_{0.025, 19}^2 = 32.852$ and $\chi_{0.975, 19}^2 = 8.907$. The confidence interval is:

$$CI = \left(\frac{(19)(25)}{32.852}, \frac{(19)(25)}{8.907} \right) = (14.46, 53.29)$$

Further Reading

You can further read the material on Interval Estimation in pages 357-363 from the ebook by Allan G. Bluman. (2009). Elementary Statistics: A step by step approach. Seventh edition, McGraw-Hill. Click Interval Estimation to read the book.

Activity 10: Mastery Quizzes

After reading pages 357-363 from the ebook, answer questions Q10, Q11, Q14 and Q15 on page 366. .

Hypothesis Testing about Parameters in AI Systems

Hypothesis is a statement or claim about a population parameter. As a result, Hypothesis Testing is a statistical method that involves making inferences or draw conclusions about a population parameter based on sample data.

Therefore, in Hypothesis Testing, you will learn how to use sample data to make inferences regarding a particular population parameter. This procedure involves formulating two competing hypotheses: the null hypothesis and the alternative hypothesis:

- The null hypothesis (H_0) represents a statement of no effect or no difference, and it is assumed to be true unless the evidence suggests otherwise.
- The alternative hypothesis (H_1 or H_a) is the statement we want to test for, suggesting that there is an effect or a difference.

Types of Hypotheses

Two-tailed Test

A two-tailed test is used when the alternative hypothesis is testing for a difference in either direction. The null and alternative hypotheses are stated as:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0$$

In this case, the rejection region is split between the two tails of the distribution.

One-tailed Test

A one-tailed test is used when the alternative hypothesis is testing for a difference in one direction only:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0 \quad (\text{right-tailed})$$

or

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu < \mu_0 \quad (\text{left-tailed})$$

In this case, the rejection region is located in only one tail of the distribution.

Steps in Hypothesis Testing

1. State the hypotheses:
 - Null hypothesis (H_0) : A statement asserting no difference or no effect.
 - Alternative hypothesis (H_1) : A statement asserting a difference or an effect.
2. Choose the significance level (α) : This is the probability of rejecting the null hypothesis when it is true. Common values are $\alpha = 0.05$ or $\alpha = 0.01$.
3. Select the appropriate test statistic: This depends on the type of data and the hypotheses being tested (e.g., z-test, t-test, χ^2 -test).
4. Formulate the decision rule: Based on the significance level and test statistic, determine the rejection region or calculate the p -value.

5. Compute the test statistic: Use sample data to compute the value of the test statistic.
6. Make the decision: Compare the test statistic to the critical value or the p-value to the significance level to accept or reject the null hypothesis.
7. State the conclusion: Interpret the results in the context of the research problem.

Errors in Hypothesis Testing

- Type I error (α) : Rejecting the null hypothesis when it is true. This is controlled by setting the significance level α .
- Type II error (β) : Failing to reject the null hypothesis when it is false. The power of the test is $1 - \beta$, which represents the probability of correctly rejecting a false null hypothesis.

Test Statistics

Z-Test

The z-test is used to test hypotheses about a population mean when the population variance is known or when the sample size is large ($n \geq 30$). The test statistic is:

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

where $\bar{x}$ is the sample mean, μ_0 is the population mean under the null hypothesis, σ is the population standard deviation, and n is the sample size.

T-Test

The t-test is used when the population variance is unknown and the sample size is small ($n < 30$). The test statistic is:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

where $\bar{x}$ is the sample mean, μ_0 is the population mean under the null hypothesis, s is the sample standard deviation, and n is the sample size. The t-distribution is used with $n - 1$ degrees of freedom.

Chi-Square Test

The χ^2 -test is used for categorical data to test hypotheses about population variances or goodness-of-fit. The test statistic for variance is:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

where s^2 is the sample variance, σ^2 is the population variance under the null hypothesis, and n is the sample size. It follows a chi-square distribution with $n - 1$ degrees of freedom.

P-Value and Significance of Parameters

The **p-value** is the probability of obtaining a test statistic at least as extreme as the one observed, assuming the null hypothesis is true.

General Understanding of P-Values

- Smaller p-values suggest stronger evidence against the null hypothesis, meaning the observed data is unlikely to have occurred if the null hypothesis were true.
- Larger p-values suggest weaker evidence against the null hypothesis, meaning the observed data is consistent with the null hypothesis.

Interpretation of P -Values In hypothesis testing

We compare the p -value to the significance level (denoted as α) to make a decision:

- If the p -value $\leq \alpha$, reject the null hypothesis (H_0). This indicates strong evidence against the null hypothesis. You reject the null hypothesis, concluding that there is a statistically significant effect or difference.
- If the p -value $> \alpha$, fail to reject the null hypothesis. This is often interpreted as weak or no evidence against the null hypothesis.
- Illustrations of Computation of P-Values

Worked Out Examples

Example 11. One-sample Z-test for Population Mean *A manufacturer claims that the average lifetime of a light bulb is 1000 hours. A sample of 50 bulbs is tested and the sample mean lifetime is found to be 990 hours with a known population standard deviation of 20 hours. Test the manufacturer's*

claim at a 5% significance level.

Solution

- *Step 1: State the hypotheses.*

$H_0 : \mu = 1000$ (The population mean lifetime is 1000 hours)

$H_1 : \mu \neq 1000$ (The population mean lifetime is different from 1000 hours)

This is a two-tailed test.

- *Step 2: Compute the test statistic.*

The formula for the Z-test statistic is:

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

where:

$\bar{x} = 990$ is the sample mean,

- $\mu_0 = 1000$ is the population mean under the null hypothesis,
- $\sigma = 20$ is the population standard deviation,
- $n = 50$ is the sample size.

Substituting the values:

$$z = \frac{990 - 1000}{\frac{20}{\sqrt{50}}} = \frac{-10}{2.828} = -3.536$$

- **Step 3: Find the p-value.**

For a two-tailed test, the p-value is:

$$p = 2 \times P(Z < -3.536)$$

From the standard normal distribution table, $P(Z < -3.536) \approx 0.0002$.

Therefore:

$$p = 2 \times 0.0002 = 0.0004$$

- **Step 4: Make a decision.**

Since the p-value (0.0004) is less than the significance level $\alpha = 0.05$, we reject the null hypothesis. There is sufficient evidence to conclude that the mean lifetime of the bulbs is different from 1000 hours.

Example 12. One-sample T-test for Population Mean

A researcher claims that the average IQ of a population is 110. A sample of 25 individuals has an average IQ of 105 with a sample standard deviation of 15. Test the researcher's claim at a 1% significance level.

Solution

- **Step 1: State the hypotheses.**

$H_0 : \mu = 110$ (The population mean IQ is 110)

$H_1 : \mu \neq 110$ (The population mean IQ is different from 110)

This is a two-tailed test.

- **Step 2: Compute the test statistic.**

The formula for the t-test statistic is:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

where:

$\bar{x} = 105$ is the sample mean,

- $\mu_0 = 110$ is the population mean under the null hypothesis,
- $s = 15$ is the sample standard deviation,
- $n = 25$ is the sample size.

Substituting the values:

$$t = \frac{105 - 110}{\frac{15}{\sqrt{25}}} = \frac{-5}{3} = -1.667$$

- **Step 3: Find the p-value.**

With $n - 1 = 24$ degrees of freedom, we look up the t-distribution table for $t = -1.667$. The one-tailed p-value is approximately $P(T < -1.667) = 0.054$. For a two-tailed test:

$$p = 2 \times 0.054 = 0.108$$

- **Step 4: Make a decision.**

Since the p-value (0.108) is greater than the significance level $\alpha = 0.01$, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the population mean IQ is different from 110.

Example 13. Two-sample Z-test for Proportions

A study is conducted to compare the proportion of males and females who prefer a certain brand of coffee. In a sample of 100 males, 60 said they prefer the brand, while in a sample of 80 females, 50 said they prefer it. Test at the 5% significance level whether the proportions are different.

Solution

- **Step 1: State the hypotheses.**

$H_0 : p_1 = p_2$ (The proportions of males and females who prefer the brand are equal),

$H_1 : p_1 \neq p_2$ (The proportions of males and females who prefer the brand are different)

- **Step 2: Compute the test statistic.**

The formula for the Z-test statistic for proportions is:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where:

- $\hat{p}_1 = \frac{60}{100} = 0.60$ is the sample proportion of males,
- $\hat{p}_2 = \frac{50}{80} = 0.625$ is the sample proportion of females,
- $p = \frac{60+50}{100+80} = \frac{110}{180} = 0.611$ is the pooled proportion,
- $n_1 = 100$ and $n_2 = 80$ are the sample sizes.

Substituting the values:

$$z = \frac{0.60 - 0.625}{\sqrt{0.611(1 - 0.611) \left(\frac{1}{100} + \frac{1}{80} \right)}} = \frac{-0.025}{0.070} = -0.357$$

- **Step 3: Find the p-value.**

For a two-tailed test, the p-value is:

$$p = 2 \times P(Z < -0.357)$$

From the standard normal distribution table, $P(Z < -0.357) \approx 0.360$.
Therefore:

$$p = 2 \times 0.360 = 0.720$$

- **Step 4: Make a decision.**

Since the p-value (0.720) is greater than the significance level $\alpha = 0.05$, we fail to reject the null hypothesis. There is no significant evidence to suggest that the proportions of males and females who prefer the brand are different.

Further Examples of Hypothesis Testing

Example 14. Suppose we want to test whether the average weight of bags of rice is different from 50 kg . We take a sample of 30 bags and find a mean weight of 49.2 kg with a known population standard deviation of 2 kg . Test at the 5% significance level.

Solution

- Null hypothesis: $H_0 : \mu = 50$.
- Alternative hypothesis: $H_1 : \mu \neq 50$ (two-tailed test).
- Significance level: $\alpha = 0.05$.
- Test statistic: $z = \frac{49.2-50}{\frac{2}{\sqrt{30}}} = -2.19$.
- Critical value: For a two-tailed test at $\alpha = 0.05$, $z_{\alpha/2} = 1.96$.
- Conclusion: Since $|z| = 2.19 > 1.96$, we reject the null hypothesis. There is evidence to suggest that the average weight of the bags is not 50 kg .

Example 15. A manufacturer claims that the average life of a certain battery is 500 hours. A random sample of 15 batteries has a mean life of 485 hours with a standard deviation of 20 hours. Test the manufacturer's claim at the 1% significance level.

Solution

- Null hypothesis: $H_0 : \mu = 500$.
- Alternative hypothesis: $H_1 : \mu \neq 500$ (two-tailed test).
- Significance level: $\alpha = 0.01$.
- Test statistic: $t = \frac{485-500}{\frac{20}{\sqrt{15}}} = -2.91$.
- Degrees of freedom: $df = 14$.
- Critical value: From the t -distribution table for $\alpha = 0.01$ and $df = 14$, $t_{\alpha/2} = 2.977$.
- Conclusion: Since $|t| = 2.91 < 2.977$, we fail to reject the null hypothesis. There is not enough evidence to reject the manufacturer's claim.